

- 2 (a) total volume of molecules negligible compared to that of containing vessel
 no intermolecular forces
 molecules in random motion
 time of collision small compared with the time between collisions
 large number of molecules
any two B2 [2]
- (b) in a real gas there is a range of velocities *or* must take the average of v^2 B1 [1]
- (c) (i) *either* $p = \frac{1}{3} \rho \langle c^2 \rangle$
or $1.0 \times 10^5 = \frac{1}{3} \times 1.2 \times \langle c^2 \rangle$ C1
- $\langle c^2 \rangle = 2.5 \times 10^5$ C1
 $c_{\text{r.m.s.}} = 500 \text{ m s}^{-1}$ A1 [3]
- (ii) $T \propto \langle c^2 \rangle$ C1
 $\langle c^2 \rangle = 2.5 \times 10^5 \times 480 / 300$
 $= 4.0 \times 10^5 \text{ m}^2 \text{ s}^{-2}$ (allow ECF from (c)(i)) A1 [2]